

$$1) a) \lim_{x \rightarrow 25} \frac{\sqrt{x} - 5}{25 - x} = \lim_{x \rightarrow 25} \frac{(\sqrt{x} - 5)(\sqrt{x} + 5)}{(25 - x)(\sqrt{x} + 5)} = \lim_{x \rightarrow 25} \frac{x - 25}{(25 - x)(\sqrt{x} + 5)} =$$

$$= \lim_{x \rightarrow 25} \frac{-(25 - x)}{(25 - x)(\sqrt{x} + 5)} = - \frac{1}{\sqrt{25} + 5} = \boxed{-\frac{1}{10}}$$

$$b) \lim_{x \rightarrow 0} \frac{x}{3 - \sqrt{9+x}} = \lim_{x \rightarrow 0} \frac{x(3 + \sqrt{9+x})}{(3 - \sqrt{9+x})(3 + \sqrt{9+x})} = \lim_{x \rightarrow 0} \frac{x(3 + \sqrt{9+x})}{9 - 9 - x} =$$

$$= \lim_{x \rightarrow 0} \frac{x(3 + \sqrt{9+x})}{-x} = -(3+3) = \boxed{-6}$$

$$2) a) y = \left(\sin \frac{x^2}{3} \right)^6 \quad y = u^m \quad y' = m \cdot u^{m-1} \cdot u'$$

$$u = \sin \frac{x^2}{3} \quad u' = \frac{2}{3} x \cdot \cos \frac{x^2}{3}$$

$$y' = 6 \left(\sin \frac{x^2}{3} \right)^5 \cdot \frac{2}{3} x \cos \frac{x^2}{3} = \boxed{4x \left(\sin \frac{x^2}{3} \right)^5 \cos \frac{x^2}{3}}$$

$$b) y = (\sin^3 2x) \cdot (\ln x)^x \quad y = s \cdot t \quad y' = s' \cdot t + s \cdot t'$$

$$s = (\sin 2x)^3 \quad s' = 3(\sin 2x)^2 \cdot 2 \cos 2x = 6(\sin 2x)^2 \cos 2x$$

$$t = (\ln x)^x \rightarrow \ln t = \ln (\ln x)^x \rightarrow \ln t = x \cdot \ln (\ln x)$$

$$\frac{t'}{t} = \ln (\ln x) + x \cdot \frac{1}{\ln x} \quad t' = \left[\ln (\ln x) + \frac{1}{\ln x} \right] (\ln x)^x$$

$$y' = 6(\sin 2x)^2 \cos 2x \cdot (\ln x)^x + (\sin 2x)^3 \cdot \left[\ln (\ln x) + \frac{1}{\ln x} \right] (\ln x)^x$$

$$c) y = \tan(x^2+x) \cdot \cos x^2 \quad y = s \cdot t \quad y' = s' \cdot t + s \cdot t'$$

$$s = \tan(x^2+x) \quad s' = \frac{2x+1}{\cos^2(x^2+x)}$$

$$t = \cos x^2 \quad t' = -2x \cdot \sin x^2$$

$$y' = \left[\frac{2x+1}{\cos^2(x^2+x)} \right] \cdot \cos x^2 + \tan(x^2+x) (-2x \sin x^2)$$